MCP, Explained and Built

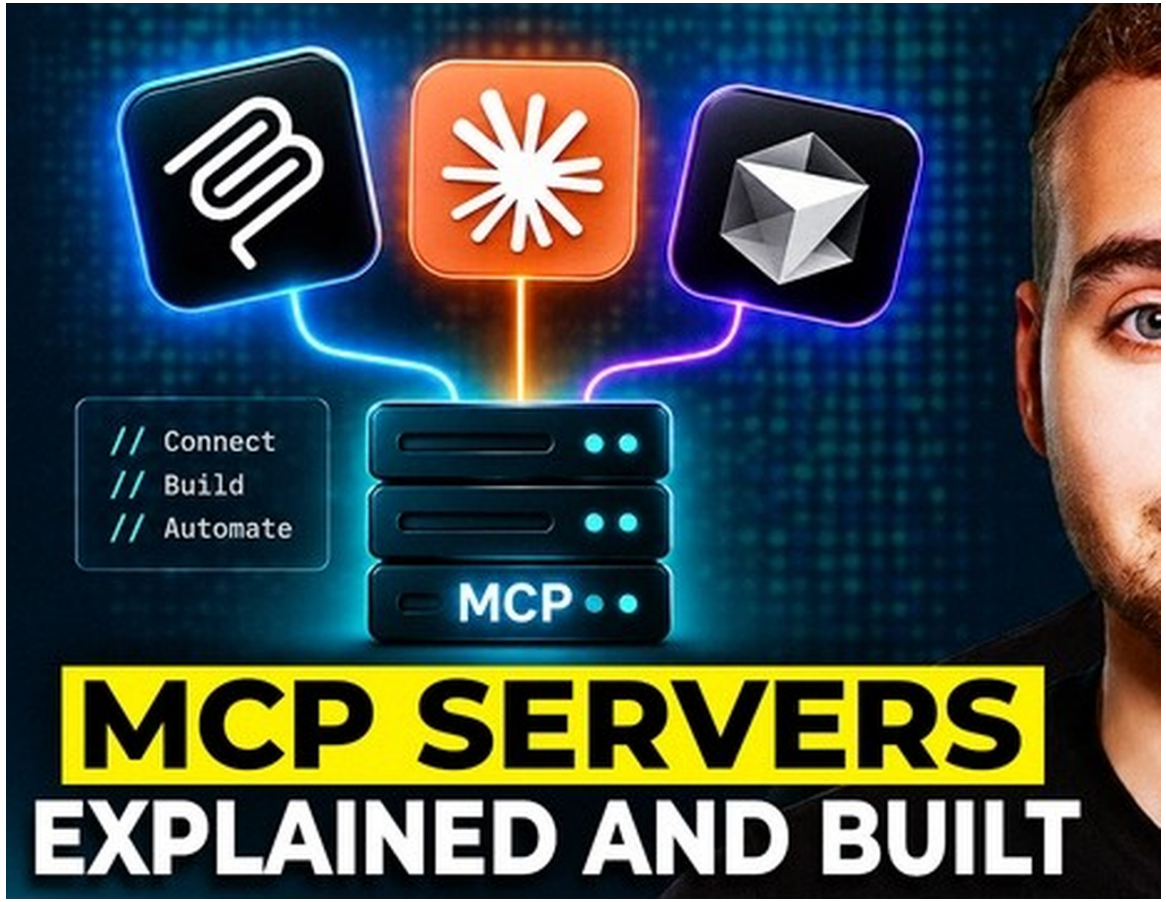
Following one tutorial, step by step

Our Reference

MCP Servers Explained

Tech With Tim, 30 min

We follow its order



Source

What We Will Cover

01 What an MCP server is

02 Local versus remote

03 Who may call your tools

04 Build a notes server

05 Add a real login

MCP Server

Gives an AI model a list of functions it may call.

A Model Alone Only Writes Text

It can't read your files

It can't check your calendar

It can't query your database

Tool

A function with a name, a description and inputs.

How a Tool Call Works

01 The model asks for a tool

By name, with its inputs

02 Your code runs it

The model never runs anything

03 The result goes back

And the model carries on

Why a Standard Helps

Before

Each app built its own

vs

MCP

One server, every app

Two Sides

MCP client	MCP server
The app the model is in	Your code
Claude Code, Cursor	Sends its tool list
Asks for the tool list	Runs the tool asked for

Local Versus Remote

	Local	Remote
Runs on	Your laptop	The internet
Reached by	Claude starts it	A URL
Used by	Only you	Anyone with the URL

Remote Means Three Questions

Who is calling?

Which app sent this

On whose behalf?

Which person it acts for

What may they do?

Read, write, or delete

Why One Shared Key Fails

Every agent looks the same

No person behind a request

One leak breaks everyone

How a Real Login Works

01 Claude calls with no token

Server says 401: log in here

02 You log in and approve

Claude gets a token

03 Claude sends the token

The server checks, then runs

What the Token Answers

Question	In the token
Who is calling?	The client id
On whose behalf?	The user id
What may they do?	The scopes
For which server?	This one only

Three Versions, One Server

v1: local, no login

v2: a URL, no login

v3: a URL with login

MCP Servers Explained + Descope (video build)

Three versions of the same "notes" MCP server, built in the order you show them on camera.

- Script: `script.md` (theory fully scripted, build + Descope as bullets)
- Brand-facing script (Google Doc): https://docs.google.com/document/d/1PCnNyEp0rB_m-_FViT_pYCKwViQmGUEISLTVpRmpPqQ/edit
- Theory slide deck: `presentation/index.html` (open locally, arrows to move, F for fullscreen) or https://clever-opera-nnya.here.now/

File	What it is	Auth
`v1_local.py`	stdio server, Claude/Cursor launch it as a subprocess	none (local only)
`v2_remote.py`	same tools over Streamable HTTP at `http://localhost:8000/mcp`	none (the problem)
`v3_auth.py`	same tools, protected by Descope, notes stored per user, scopes per tool	Descope OAuth 2.1
`test_client.py`	terminal client; `--oauth` triggers the browser login flow	
`notes_db.py`	SQLite helper shared by all three	

Versions pinned on purpose: `fastmcp<4` and `mcp<2`, because `descope-mcp` 0.1.0 still imports `FastMCP` from the old `mcp.server` path and breaks on `mcp` 2.x. Already done in `pyproject.toml`.

Setup

Source

v1_local.py

```
from fastmcp import FastMCP
mcp = FastMCP("notes")

@mcp.tool()
def add_a_note(text: str):
    """Save a new note."""
    return add_note("local", text)
mcp.run()
```

What the Model Reads

The docstring is the description

Type hints say what to pass

Write both on every tool

Connect it locally

```
$ claude mcp add notes \
    -- uv run v1_local.py

> What can my notes
  server do?
```

v2_remote.py

```
mcp.run(transport="http",
        port=8000)
```

```
$ claude mcp add \
    --transport http \
    notes \
    http://localhost:8000/mcp
```

Set Up the Login Service

01 Add your server

Its URL, ending in /mcp

02 Add two scopes

notes:read and notes:write

03 Copy the config URL

Into your .env file

v3_auth.py: the gate

```
auth = DescopeProvider(
    config_url=CONFIG_URL,
    base_url=BASE_URL,
    scopes_supported=[
        "notes:read",
        "notes:write"])
mcp = FastMCP("notes", auth=auth)
```

v3_auth.py: each tool

```
@mcp.tool()
def add_a_note(text: str):
    user, _ = current_user(
        "notes:write")
    return add_note(user, text)
```

Test It With Two Accounts

01 Run /mcp and sign in

Add "Tim is the best"

02 Sign in as someone else

A second account

03 List the notes

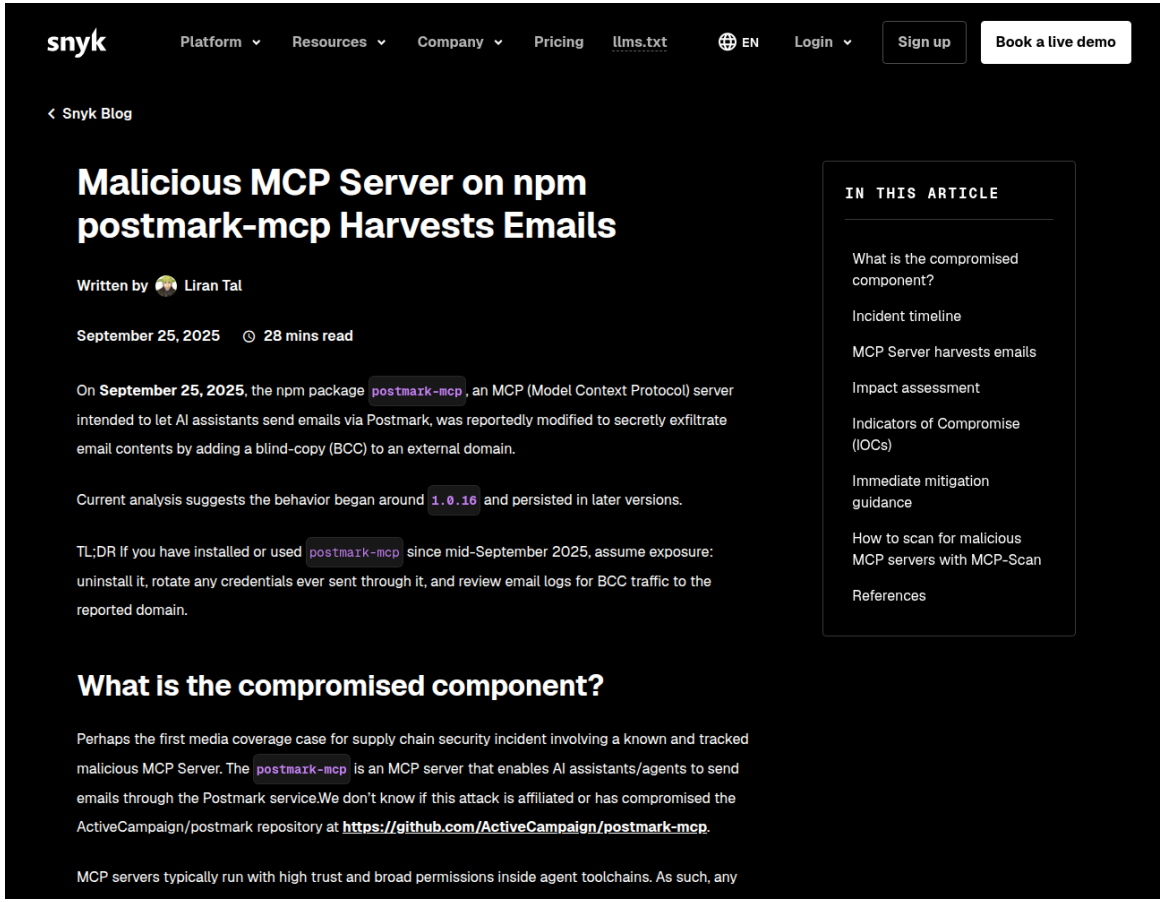
They see none

The Other Direction

A server can attack you

postmark-mcp, Sept 2025

It copied every email



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Snyk Blog

Malicious MCP Server on npm postmark-mcp Harvests Emails

Written by Liran Tal

September 25, 2025 28 mins read

On **September 25, 2025**, the npm package `postmark-mcp`, an MCP (Model Context Protocol) server intended to let AI assistants send emails via Postmark, was reportedly modified to secretly exfiltrate email contents by adding a blind-copy (BCC) to an external domain.

Current analysis suggests the behavior began around `1.0.16` and persisted in later versions.

TL;DR If you have installed or used `postmark-mcp` since mid-September 2025, assume exposure: uninstall it, rotate any credentials ever sent through it, and review email logs for BCC traffic to the reported domain.

IN THIS ARTICLE

- What is the compromised component?
- Incident timeline
- MCP Server harvests emails
- Impact assessment
- Indicators of Compromise (IOCs)
- Immediate mitigation guidance
- How to scan for malicious MCP servers with MCP-Scan
- References

What is the compromised component?

Perhaps the first media coverage case for supply chain security incident involving a known and tracked malicious MCP Server. The `postmark-mcp` is an MCP server that enables AI assistants/agents to send emails through the Postmark service.We don't know if this attack is affiliated or has compromised the ActiveCampaign/postmark repository at **https://github.com/ActiveCampaign/postmark-mcp**.

MCP servers typically run with high trust and broad permissions inside agent toolchains. As such, any

Source

Sources

Part	Taken from
The tutorial	youtube.com, Tech With Tim
Its code	github.com/techwithtim
Login rules	modelcontextprotocol.io
Claude commands	code.claude.com/docs
The attack	snyk.io/blog

Before You Build or Connect

A remote server needs a login

Give each tool one scope

Connect only servers you trust

Pin the version you checked